Neural Networks 206 (2027) 109829

Contents lists available at ScienceDirect

Neural Networks

journal homepage: www.elsevier.com/locate/neunet

Full Length Article

OTSNet: A unified observation-thinking-spelling network for robust scene text recognition

Lixu Sun, Jian Peng, Nurmemet Yolwas*, Wushour Silamu

School of Computer Science and Technology, Xinjiang University, Urumqi, 830046, China

ARTICLE INFO

Keywords:
Scene text recognition
Differential attention
Semantic quantization
Cross-modal fusion

ABSTRACT

Scene Text Recognition (STR) remains challenging because many existing methods decouple visual and linguistic optimization, often relying on large-scale pre-training to bridge the visual-semantic gap. Inspired by human reading behavior, we propose OTSNet, a unified "Observation-Thinking-Spelling" framework. OTSNet contains three coordinated stages: (1) Observation: a Dual Attention Macaron Encoder (DAME) applies differential attention as an attention sharpening mechanism to emphasize glyph-relevant regions while suppressing background noise; (2) Thinking: a Position-Aware Module (PAM) and a Semantic Quantizer (SQ) transform visual features into discrete glyph-level semantic units; and (3) Spelling: a Multi-Modal Collaborative Verifier (MMCV) performs autoregressive decoding by jointly exploiting visual features, quantized semantic units, and character embeddings. Unlike decoupled pipelines, OTSNet aligns visual and semantic representations without gradient blocking and supports end-to-end training from scratch. Without pre-training, OTSNet achieves 83.5% on Union14M-L and 79.1% on OST, attaining the highest aggregate accuracy on both robustness-oriented benchmark suites among the methods compared. These results demonstrate strong robustness under complex and irregular scene text conditions.

1. Introduction

Scene Text Recognition (STR), which aims to recognize text in complex natural images, is a core problem in computer vision (Fu et al., 2023). Real-world STR remains difficult due to irregular layouts (e.g., curved and perspective text) and severe degradations such as occlusion, blur, and background clutter. Early STR methods mainly focused on character-level segmentation and classification, with limited semantic modeling.

Recent advances have improved performance by incorporating Language Models (LMs) (Qiao et al., 2020). A representative design, exemplified by Fang et al. (2021), decouples the Visual Model (VM) and LM, and blocks gradients between them to stabilize iterative optimization (Fig. 1(a)). However, this rigid decoupling weakens bidirectional interaction between visual evidence and semantic priors, which can lead to erroneous predictions under ambiguous visual conditions.

To narrow this gap, recent methods like BUSNet (Wei et al., 2024) attempt tighter vision-language coupling through unified representation learning (Fig. 1(b)). However, to stabilize such complex multi-modal interactions, these architectures often rely on staged pre-training and detach the gradients of intermediate text features before final decoding. Consequently, the optimization between visual perception and semantic reasoning remains fundamentally disjoint. This structural compromise forces models to overly depend on linguistic priors, which can lead to visually inconsistent predictions when encountering heavily degraded or ambiguous text. In contrast, human reading is a straightforward and adaptable process: when facing an ambiguous string, people naturally reduce their reliance on semantic guessing and accurately transcribe the text simply by observing the character strokes and shapes. Motivated by this intuitive behavior, we restructure STR into a simple, fully differentiable "Observation-Thinking-Spelling" pipeline that requires neither gradient blocking nor pre-training.

Specifically, we translate this human reading behavior into three corresponding computational stages: visual observation (Kastner & Ungerleider, 2000), semantic abstraction (Zheng et al., 2024), and sequence decoding (Yang et al., 2024). To instantiate this pipeline, we propose OTSNet (Fig. 2), which aligns its architecture with these three stages: (1) For the Observation stage, the Dual Attention Macaron Encoder (DAME) acts as an attention sharpening mechanism, alternating standard and differential self-attention to extract discriminative visual evidence and improve feature selectivity in cluttered scenes. (2) For the Thinking stage, a Position-Aware Module (PAM) aligns patch features with fixed character positions, while a Semantic Quantizer (SQ) converts these aligned features into discrete glyph-level semantic units through differentiable codebook learning. (3) Finally, for the Spelling stage, the Multi-Modal Collaborative Verifier (MMCV) performs autoregressive multimodal

* Corresponding author.
E-mail addresses: sunlixu@stu.xju.edu.cn (L. Sun), pengj@stu.xju.edu.cn (J. Peng), nurmemet@xju.edu.cn (N. Yolwas), wushour@xju.edu.cn (W. Silamu).

https://doi.org/10.1016/j.neunet.2026.109829
Received 8 March 2026; Received in revised form 7 September 2026; Accepted 29 October 2026
Available online 2 November 2026
0893-6080/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

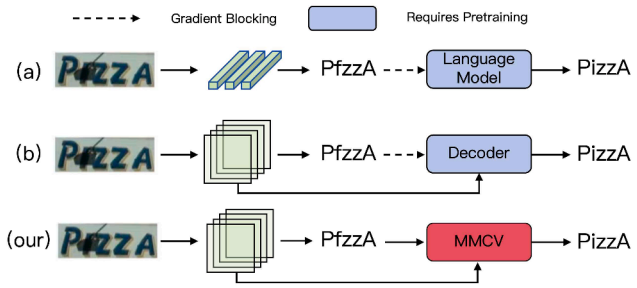


Fig. 1. Comparison of STR frameworks. (a) ABINet: decoupled LM with gradient blocking and heavy pre-training. (b) BUSNet: visually enhanced decoding but still uses gradient blocking and pre-training. (c) OTSNet (ours): fully differentiable visual-semantic fusion, enabling end-to-end training from scratch.

decoding by jointly exploiting visual features, discrete semantic units, and character embeddings. Together, these three stages train end-to-end from scratch, with no gradient blocking between the visual and semantic branches.

The main contributions of this paper are summarized as follows:

1. We propose OTSNet, a cognition-inspired STR framework that organizes visual encoding, glyph-level semantic abstraction, and multi-modal decoding into the three functional stages of Observation, Thinking, and Spelling.
2. We design DAME, a hybrid encoder that interleaves standard and differential self-attention to suppress background noise and sharpen discriminative text features.
3. We introduce a Semantic Quantizer (SQ) that transforms PAM's position-aware visual features into discrete glyph-level semantic units, forming an intermediate semantic abstraction between spatial localization and multi-modal verification.
4. OTSNet achieves state-of-the-art robustness from scratch, attaining the best Union14M-L average of 83.5% and the best OST accuracy of 79.1% among the methods compared.

2. Related work

2.1. Scene text recognition

Deep learning-based Scene Text Recognition (STR) has gradually shifted from segmentation-based pipelines, which require costly character-level annotations, to sequence recognition models. Early sequence methods mainly focused on extracting and aligning visual features. CTC-based approaches typically combined convolutional encoders with recurrent networks for sequence modeling. ASTER (Shi et al., 2018) introduced spatial rectification followed by attention-based decoding, whereas ViTSTR (Atienza, 2021) adopted a pure Transformer architecture for STR. Recent studies have continued to improve the visual branch. LMIM (Zhang et al., 2025) introduces linguistic context into masked image modeling for visual pre-training, Cascaded-Transformers (Ozkan et al., 2025) progressively reduces visual tokens to improve encoding efficiency, and SVTRv2 (Du et al., 2025) strengthens CTC-based recognition through multi-size resizing, feature rearrangement, and semantic guidance.

When text images are severely degraded, however, visual evidence alone may not be sufficient. Methods such as SRN (Yu et al., 2020), PIMNet (Qiao et al., 2021), and ABINet (Fang et al., 2021) therefore introduce semantic reasoning to refine visual predictions. ABINet, for example, separates visual and language modeling and blocks gradients between the two branches during training. This separation allows the two components to be trained independently, but also limits direct feature interaction between them.

Later studies moved toward closer visual-semantic interaction. PARSeq (Bautista & Atienza, 2022) uses permutation language modeling for autoregressive recognition, while MATRN (Na et al., 2022) enhances visual and semantic features through cross-modal interaction. VOLTER (Li et al., 2024) explores visual collaboration with a dual-stream design, and BUSNet (Wei et al., 2024) performs balanced vision-language reasoning in a unified framework. CDistNet (Zheng et al., 2024) approaches robust recognition from a different perspective by modeling character relationships across multiple feature domains.

These methods show that end-to-end training and visual-semantic interaction are not new to STR. However, the representation passed between visual encoding and semantic decoding has received less attention. Most existing methods rely on continuous intermediate features or directly refine character predictions, rather than constructing explicit glyph-level discrete units. Strong linguistic context can then dominate uncertain visual evidence in heavily degraded images or arbitrary character strings, producing plausible predictions that do not match the observed glyphs. We therefore explore a glyph-level discrete representation between the position-aligned visual features and the subsequent semantic decoder.

2.2. Discrete representation learning

Discrete representation learning converts continuous features into a finite set of learned codes. A representative method, VQ-VAE (van den Oord et al., 2017), assigns encoder outputs to codebook entries and uses a straight-through estimator to optimize the discrete latent space for image reconstruction.

Recent multimodal studies have also used discrete intermediate representations. TokenFlow (Qu et al., 2025) aligns semantic and pixel-level codebooks to support both multimodal understanding and image generation. In a perception-before-reasoning design, AURORA (Bigverdi et al., 2025) uses a VQ-VAE to convert intermediate perceptual outputs, such as depth maps, into perception tokens that support subsequent visual reasoning.

In OTSNet, PAM first produces one position-aware feature for each fixed character slot. SQ uses these features to select one glyph embedding for each slot under character-level supervision, and MMCV combines the selected embeddings with the visual features and autoregressive character embeddings for recognition.

2.3. Visual feature enhancement methods

A complementary line of work improves STR robustness by enhancing visual representations under challenging imaging conditions and irregular text patterns. These methods generally fall into two categories: image-level enhancement and architecture-driven visual modeling.

Image-level methods aim to reduce the influence of degradation before feature extraction. For instance, Luo et al. (2021) used adversarial learning to separate textual content from style variations. Meanwhile, Li et al. (2021) introduced character-aware sampling and rectification to alleviate character distortion, while Wu et al. (2022) developed a two-level rectification attention network for irregular text recognition.

Alternatively, architecture-driven methods handle irregular patterns directly in the feature space. Lee et al. (2020) employed 2D self-attention to capture spatial dependencies for arbitrarily shaped text. Similarly, Cheng et al. (2018) combined multi-directional feature sequences with a filtering gate to recognize arbitrarily oriented text. More recently, SVTR (Du et al., 2022) mixed local and global receptive fields to strengthen visual feature extraction. However, standard self-attention may still assign non-trivial responses to background regions in cluttered scenes, weakening the distinction between glyph features and surrounding noise.

Several attention mechanisms have been explored to reduce irrelevant responses. Top-k attention restricts each query to its strongest attention scores (Gupta et al., 2021), while gated cascade attention refines character features through spatial and channel attention (Wang et al., 2019). Contrastive attention further distinguishes the most

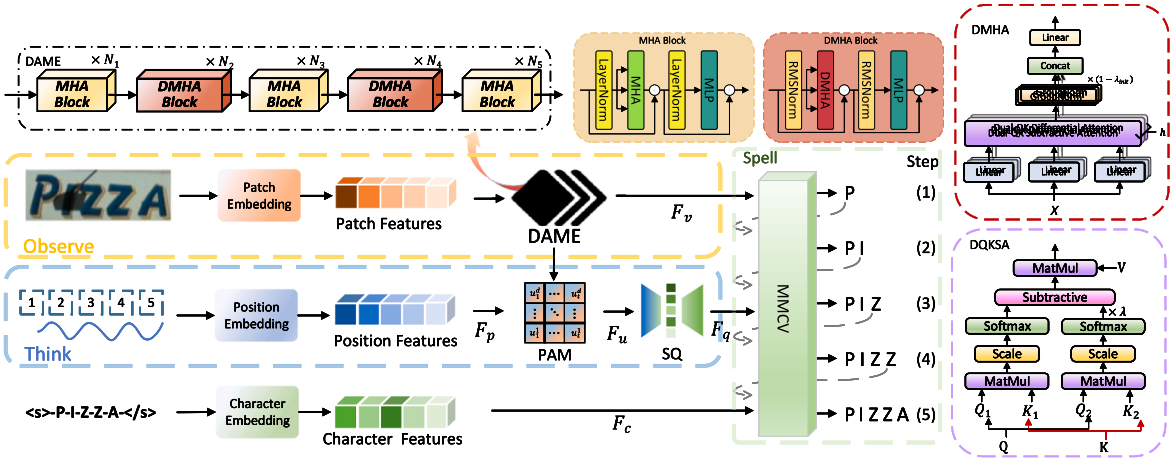


Fig. 2. An overview of OTSNet. DAME extracts visual features from the input image, PAM aligns them with fixed character positions, and SQ converts the position-aware features into discrete glyph-level semantic features. MMCV then combines the visual, semantic, and character features to produce the final recognition output.

relevant image regions from the least relevant ones during semantic refinement (Zhou et al., 2023). Unlike these sparse, gated, or contrastive designs, DMHA suppresses background responses through attention differencing. It subtracts a scaled secondary attention map from the primary one, without using a top-k threshold or an additional contrastive objective. Based on this mechanism, DAME alternates MHSA and DMHA to preserve global context while progressively sharpening glyph-related visual features.

3. Methodology

OTSNet follows the Observation-Thinking-Spelling pipeline shown in Fig. 2. In Observation, DAME extracts denoised and discriminative visual features from image patches. In Thinking, PAM aligns visual tokens with character positions, and SQ converts them into discrete glyph-level semantic units. In Spelling, MMCV fuses visual, semantic, and character features for final prediction.

3.1. Observation

Transformer-based models capture global context effectively, but standard self-attention can suffer from attention dispersion in cluttered scenes. This behavior assigns non-trivial attention weights to background regions and weakens glyph discriminability.

To address this issue, DAME interleaves standard MHSA and DMHA in a Macaron-style hierarchy. This alternating structure provides progressive optimization: global context modeling (MHSA) followed by local discriminative refinement (DMHA).

MHSA follows the standard Transformer residual structure. DMHA implements differential attention (Ye et al., 2024) through a Dual-QK Subtractive Attention mechanism (DQKSA). Instead of relying on standard attention alone, this mechanism acts as an attention differencing and sharpening mechanism. By subtracting a scaled secondary attention map from the primary one, it amplifies discriminative patterns (e.g., character strokes) while dynamically suppressing redundant background noise.

Given input features $X \in \mathbb{R}^{B\times N\times D}$, DMHA first generates dual independent query-key projections:

$$Q_1 = XW_{q_1}, \quad Q_2 = XW_{q_2}, \tag{1}$$

$$K_1 = XW_{k_1}, \quad K_2 = XW_{k_2}, \quad V = XW_v, \tag{2}$$

where $Q_1, Q_2, K_1, K_2 \in \mathbb{R}^{B\times N\times D/2}$ and $V \in \mathbb{R}^{B\times N\times D}$. Two independent attention maps are then computed:

$$A_1 = \mathrm{Softmax}(Q_1 K_1^\top / \sqrt{d}), \tag{3}$$

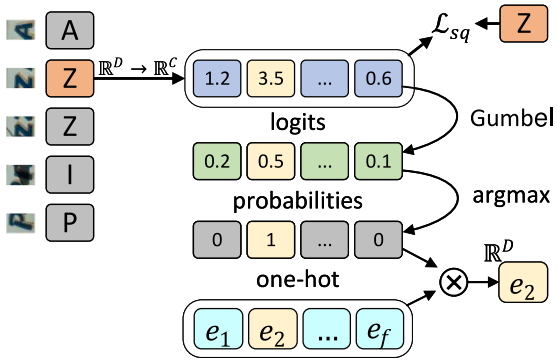


Fig. 3. Schematic diagram of the Semantic Quantizer (SQ) workflow.

$$A_2 = \mathrm{Softmax}(Q_2 K_2^\top / \sqrt{d}). \tag{4}$$

The attended output is obtained by a subtractive operation:

$$\mathrm{Output} = (A_1 - \lambda \cdot A_2)V, \tag{5}$$

where the dynamic trade-off parameter λ is defined as:

$$\lambda = \exp(\lambda_{q_1}^\top \lambda_{k_1}) - \exp(\lambda_{q_2}^\top \lambda_{k_2}) + \lambda_{\mathrm{init}}. \tag{6}$$

Here, $\lambda_{q_1}, \lambda_{k_1}, \lambda_{q_2}, \lambda_{k_2} \in \mathbb{R}^D$ are independent learnable vectors used only to parameterize the scalar trade-off coefficient λ, and λ_{init} is a fixed scalar initialization.

3.2. Thinking

The Observation stage produces patch-wise visual features $F_v \in \mathbb{R}^{B\times N\times D}$, which are not directly aligned with the output character positions. To obtain position-wise features, we introduce T fixed character slots.

First, sinusoidal embeddings are used to generate positional features $F_p \in \mathbb{R}^{B\times T\times D}$. PAM then uses F_p as queries and F_v as keys and values to collect visual information for each character slot:

$$F_u = \mathrm{MHCA}(F_p, F_v, F_v). \tag{7}$$

The output $F_u \in \mathbb{R}^{B\times T\times D}$ contains the position-aware visual features of the T character slots.

Next, SQ converts F_u into discrete glyph-level semantic features, as shown in Fig. 3. A learnable projection $\Phi : \mathbb{R}^D \to \mathbb{R}^C$ first produces the assignment logits:

$$Q = \Phi(F_u; \theta_\Phi), \quad Q \in \mathbb{R}^{B\times T\times C}, \tag{8}$$

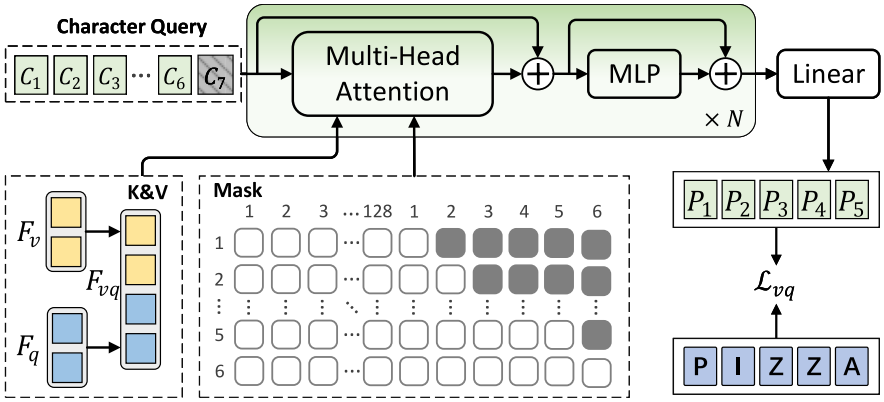


Fig. 4. Architecture of MMCV with its block-wise attention mask.

where C equals the alphabet size V (e.g., $C = 96$). For each character slot, Q provides the scores used to select one semantic unit. These logits are directly supervised by L_{sq} (Section 3.4) and are used to construct the intermediate semantic features F_q, rather than as the final recognition output.

To obtain a discrete assignment while preserving end-to-end optimization, SQ adopts hard Gumbel-Softmax (Jang et al., 2017). For the tth character slot, the continuous assignment is computed as

$$\pi_i^{(t)} = \frac{\exp\left((Q_i^{(t)} + G_i^{(t)})/\tau\right)}{\sum_{j=1}^{C} \exp\left((Q_j^{(t)} + G_j^{(t)})/\tau\right)}. \tag{9}$$

where $G_i^{(t)} \sim \text{Gumbel}(0, 1)$ and τ denotes the temperature parameter. The corresponding one-hot assignment and its straight-through form are defined as

$$z^{(t)} = \text{onehot}\left(\arg\max_i \pi_i^{(t)}\right), \tag{10}$$

$$\hat{z}^{(t)} = z^{(t)} - \text{sg}\left(\pi^{(t)}\right) + \pi^{(t)}, \tag{11}$$

where $\text{sg}(\cdot)$ denotes stop-gradient. In the forward pass, $\hat{z}^{(t)}$ has the same one-hot value as $z^{(t)}$, while the gradient is computed through the continuous assignment $\pi^{(t)}$ during back-propagation.

Finally, the selected assignment is mapped to a semantic feature through a learnable embedding table $\mathbf{E} \in \mathbb{R}^{C\times D}$:

$$F_q^{(t)} = \hat{z}^{(t)}\mathbf{E}, \quad t = 1, \ldots, T. \tag{12}$$

The resulting $F_q \in \mathbb{R}^{B\times T\times D}$ passes one selected glyph embedding for each character slot to MMCV, rather than a probability-weighted mixture of multiple embeddings. Unlike vector quantization methods that optimise a reconstruction objective, SQ's embedding table $\mathbf{E}$ is supervised directly by L_{sq}, so each entry encodes a glyph prototype rather than a general visual code. The Gumbel perturbation in $\pi^{(t)}$ allows the model to sample from the distribution over competing character assignments during training, rather than always committing to the current argmax. This maintains gradient flow across all entries of $\mathbf{E}$ throughout training.

3.3. Spelling

In the Spelling stage, MMCV (Fig. 4) performs multimodal autoregressive decoding by jointly exploiting visual features F_v, quantized semantic features F_q, and character embeddings F_c.

For a target sequence $Y = (y_1, \ldots, y_T)$, the character-side input during training is the right-shifted sequence $\bar{Y} = (\langle\text{BOS}\rangle, y_1, \ldots, y_{T-1})$, and $F_c = \text{Emb}(\bar{Y})$. Thus, the query at position i is constructed from $\langle\text{BOS}\rangle$ when $i = 1$ and from y_{i-1} otherwise; the current target y_i and subsequent characters are not included in this query. During inference, y_{i-1} is replaced by the character predicted at the preceding decoding step.

For clarity, let N_v denote the number of visual tokens in this subsection. We construct a joint multimodal memory by concatenating the visual and semantic features:

$$F_{vq} = \text{Concat}(F_v, F_q) \in \mathbb{R}^{B\times(N_v+T)\times D}. \tag{13}$$

MMCV then performs autoregressive decoding through masked cross-attention, where character features F_c are queries (Q), and multimodal features F_{vq} are keys and values (K, V):

$$\text{CA}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M^{\text{attn}}\right)V. \tag{14}$$

The attention mask is divided into visual and semantic blocks:

$$M^{\text{attn}} = \left[\mathbf{0}_{T\times N_v} \mid M^{\text{sem}}\right] \in \mathbb{R}^{T\times(N_v+T)}, \tag{15}$$

$$M^{\text{sem}}_{ik} = \begin{cases} 0, & k \le i, \\ -\infty, & k > i, \end{cases} \quad i, k \in \{1, \ldots, T\}. \tag{16}$$

The visual block remains unmasked, so every decoding step can attend to all N_v visual tokens, whereas the causal lower-triangular semantic block restricts the ith query to semantic positions up to i. The semantic memory $F_q = \text{SQ}(\text{PAM}(F_v))$ is inferred from the input image and does not contain ground-truth character embeddings. Thus, right shifting prevents target leakage through the character query, while M^{sem} controls the position-wise visibility of the image-derived semantic memory. The mask determines which memory tokens are visible rather than assigning a fixed weight to either modality; their relative contributions are learned through the attention scores over the visible keys. At the cross-attention level, removing the semantic segment gives $F_{vq} = F_v$ and $M^{\text{attn}} = \mathbf{0}_{T\times N_v}$, recovering the visual-only cross-attention used by conventional STR decoders (e.g., NRTR Sheng et al., 2019). MMCV retains the same scaled dot-product attention operation but augments the visual memory with position-aligned semantic tokens and applies segment-specific visibility constraints.

3.4. Optimization objectives

OTSNet is optimized end-to-end using a multi-task cross-entropy objective. The semantic quantization loss L_{sq} supervises the position-wise logits produced in the Thinking stage:

$$L_{\text{sq}} = \frac{1}{T}\sum_{t=1}^{T} \text{CrossEntropy}(Q^{(t)}, Y^{(t)}). \tag{17}$$

The primary autoregressive recognition loss L_{rec} is computed from the final character predictions produced by MMCV. The overall objective is

$$L = L_{\text{rec}} + \alpha_1 L_{\text{sq}}, \tag{18}$$

where α_1 is set to 0.3. Thus, L_{sq} provides intermediate character-level supervision, while L_{rec} supervises the final sequence prediction.

4. Experiments

4.1. Datasets and implementation details

We evaluate OTSNet on benchmarks covering a full spectrum of scene text recognition (STR) scenarios:

1. **General benchmarks:** ICDAR 2013 (IC13) (Karatzas et al., 2013), Street View Text (SVT) (Wang et al., 2011), IIIT5K (Mishra et al., 2012), ICDAR 2015 (IC15) (Karatzas et al., 2015), Street View Text Perspective (SVTP) (Phan et al., 2013), and CUTE80 (Risnumawan et al., 2014). Following standard evaluation protocols, we adopt the widely used versions with 857 and 1811 test images for IC13 and IC15, respectively.
2. **Challenging robustness benchmark:** Union14M-L (Jiang et al., 2023), a comprehensive test set consisting of seven subsets: Curved (Cur), Multi-Oriented (M-O), Artistic Font (Art), Contextless (Con), Salient (Sal), Multi-Word (M-W), and General (Gen).
3. **Occlusion-specific benchmark:** Occluded Scene Text (OST) (Wang et al., 2021), which includes a mild-occlusion subset (OSTw) and a heavy-occlusion subset (OSTh) for robustness evaluation under occluded scenarios.

Compared to general STR benchmarks, Union14M-L and OST contain more severe geometric distortion, background clutter, character occlusion, and layout variation, which are more aligned with real-world industrial scenarios. We thus report results on all datasets, with a particular focus on model performance under these challenging settings.

For training, we use the real-world Union14M dataset. To avoid data leakage caused by sample overlap between the original training and test splits, we adopt U14M-Filter (Du et al., 2025), which removes all overlapping samples. Unless otherwise specified, all models are trained from scratch on U14M-Filter without any synthetic data pre-training.

Implementation Details: We train OTSNet end-to-end using the AdamW optimizer with a weight decay of 0.05 and a batch size of 512 on two NVIDIA A40 GPUs, with automatic mixed-precision training enabled. A one-cycle learning rate schedule is adopted with a maximum learning rate of 5×10^{-4}, including 1.5 warm-up epochs within a total of 20 training epochs. All input images are uniformly resized to 32×128 pixels. Training data augmentation includes random rotation within $\pm 15°$, random perspective distortion, motion blur, and Gaussian noise. The DAME module adopts a 12-layer Macaron-style configuration ($N_1 = 2, N_2 = 1, N_3 = 6, N_4 = 1, N_5 = 2$) with $\lambda_{\text{init}} = 0.05$, and MMCV consists of three stacked attention–MLP blocks. In our implementation, the character vocabulary contains 96 classes, and the maximum decoding text length is set to 25.

We also compare DMHA with sparse, background-aware, and contrastive attention. Sparse attention retains the top 25% of attention scores per query and head ($k = 32$) before softmax. Background-aware attention modulates each token's MHSA output using a learned 384-to-1 sigmoid gate computed from its input. Contrastive attention uses two independent query–key branches, multiplying the primary softmax weights by one minus the sigmoid of the secondary scaled attention scores, followed by row-wise normalization. All variants retain the OTSNet training objective without additional auxiliary losses.

To ensure a fair inference speed evaluation, the Frames Per Second (FPS) is uniformly measured on a single NVIDIA 1080Ti GPU with a batch size of 1.

4.2. Comparison with state-of-the-art

We benchmark OTSNet against 18 state-of-the-art (SOTA) methods trained on U14M-Filter. As reported in Table 1, OTSNet obtains the highest Union14M-L average of 83.50% and the highest OST accuracy of 79.1% among the compared methods.

Challenging Benchmarks: OTSNet sets a new state-of-the-art on Union14M-L with an average accuracy of **83.5%**. On geometry-sensitive

Table 1
Accuracy comparison of OTSNet with other methods on different benchmarks.

IIIT5K	SVT	IC13	IC15	SVTP	CUTE80	||	Curve	Multi-Oriented	Artistic	Contextless	Salient	Multi-Words	General					
Method	Venue	Common Benchmarks					Union14M Benchmarks								Avg	OST	Size	FPS
CRNN (Shi et al., 2016)	TPAMI16	82.9	81.6	91.1	69.4	70.0	65.5	76.75	48.1	13.0	51.2	62.3	41.4	60.4	68.2	49.24	58.0	**16.20**	172
ASTER (Shi et al., 2018)	TPAMI19	96.1	93.0	94.9	86.1	87.9	92.0	91.68	70.9	82.2	56.7	62.9	73.9	58.5	76.3	68.75	61.9	19.04	67.1
NRTR (Sheng et al., 2019)	ICDAR19	98.1	96.8	97.8	88.9	93.3	94.4	94.89	67.9	42.4	66.5	73.6	66.4	77.2	78.3	67.46	74.8	44.26	17.3
DAN (Wang et al., 2020)	AAAI20	97.5	94.7	96.5	87.1	89.1	94.4	93.24	74.9	63.3	63.4	70.6	70.2	71.1	76.8	70.05	61.8	27.71	99.0
SRN (Yu et al., 2020)	CVPR20	97.2	96.3	97.5	87.9	90.9	96.9	94.45	78.1	63.2	66.3	65.3	71.4	58.3	76.5	68.43	64.6	51.70	67.1
SEED (Qiao et al., 2020)	CVPR20	96.5	93.2	94.2	87.5	88.7	93.4	92.24	69.1	80.9	56.9	63.9	73.4	61.3	76.5	68.87	62.6	23.95	65.4
RoScanner (Yue et al., 2020)	ECCV20	98.5	95.8	97.7	88.2	90.1	**97.6**	94.65	79.4	68.1	70.5	79.6	71.6	82.5	80.8	76.08	68.6	47.98	64.1
ABINet[a,b] (Fang et al., 2021)	CVPR21	98.5	98.1	97.7	90.1	94.1	96.5	95.83	80.4	69.0	71.7	74.7	77.6	76.8	79.8	75.72	75.0	36.86	73.0
VisionLAN (Wang et al., 2021)	ICCV21	98.2	95.8	97.1	88.6	91.2	96.2	94.50	79.6	71.4	67.9	73.7	76.1	73.9	79.1	74.53	66.4	32.88	93.5
PARSeq[b] (Bautista & Atienza, 2022)	ECCV22	98.1	96.3	97.9	89.2	92.1	96.2	95.00	83.5	83.2	70.4	78.3	80.7	78.9	83.1	79.73	78.1	23.83	19
MATRN[a,b] (Na et al., 2022)	ECCV22	**98.8**	**98.3**	97.9	**90.3**	95.2	97.2	**96.29**	82.2	73.0	73.4	76.9	79.4	77.4	81.0	77.62	77.8	44.34	46.9
MGP-STR (Wang et al., 2022)	ECCV22	97.9	97.8	97.1	89.6	95.2	96.9	95.74	85.2	83.7	72.6	75.1	79.8	71.1	83.1	78.65	78.7	148.00	120
SVTR (Du et al., 2022)	IJCAI22	98.0	97.1	97.3	88.6	90.7	95.8	94.58	76.2	44.5	67.8	78.7	75.2	77.9	77.8	71.17	69.6	18.09	161
LPV-B (Zhang et al., 2023)	IJCAI23	98.6	97.8	98.1	89.8	93.6	**97.6**	95.93	86.2	78.7	75.8	80.2	82.9	81.6	82.9	81.20	77.7	30.54	82.6
LISTER [a] (Cheng et al., 2023)	ICCV23	**98.8**	97.5	**98.6**	90.0	94.4	96.9	95.48	78.7	68.8	73.7	**81.6**	74.8	82.4	83.5	77.64	77.1	51.11	44.6
CDistNet (Zheng et al., 2024)	IJCV24	98.7	97.1	97.8	89.6	93.5	96.9	95.59	81.7	77.1	72.6	78.2	79.9	79.7	81.1	78.62	71.8	43.32	15.9
BUSNet[a,b] (Wei et al., 2024)	AAAI24	98.3	98.1	97.8	90.2	**95.3**	96.5	96.06	83.0	82.3	70.8	77.9	78.8	71.2	82.6	78.10	78.7	32.10	83.3
OTE (Xu et al., 2024)	CVPR24	98.6	96.6	98.0	90.1	94.0	97.2	95.74	86.0	75.8	74.6	74.7	81.0	65.3	82.3	77.09	77.8	20.28	55.2
SVTRv2[b] (Du et al., 2025)	ICCV25	98.1	96.1	98.1	89.6	91.0	**97.6**	95.08	86.6	85.2	74.7	81.0	81.6	82.6	82.7	82.06	71.9	19.80	143
OTSNet	-	98.1	96.6	98.4	90.2	94.4	97.2	95.82	**87.2**	**87.7**	**76.7**	**81.6**	**83.6**	**82.9**	**84.8**	**83.50**	**79.1**	28.6	79.2

All models are trained on U14M-Filter. Size denotes the number of parameters (M). FPS is uniformly measured on a single NVIDIA 1080Ti GPU. [a] denotes models that require pre-training.
[b] indicates reproduced results. Bold values denote the best accuracy in each column.



Fig. 5. Qualitative recognition results where BUSNet fails but OTSNet succeeds. For each image, the right-side annotation comprises: (1) Ground-truth text (top), (2) BUSNet predictions with red-colored error highlighting (middle), and (3) Our OTSNet predictions (bottom).

subsets (Curved and Multi-Oriented), OTSNet clearly outperforms representative rectification-based methods (e.g., SVTR, CDistNet), indicating that the Thinking stage handles geometric deformation effectively. OTSNet also ranks first on OST with 79.1%, demonstrating strong robustness under heavy occlusion.

Comparison with BUSNet: Relative to BUSNet, OTSNet improves all Union14M-L subsets, with the largest gains of +11.7% on Multi-Word and +5.9% on Artistic. Qualitative results (Fig. 5) further show that OTSNet handles heavily distorted fonts and cluttered backgrounds more reliably.

Cross-benchmark performance: OTSNet achieves an average accuracy of 95.82% on the six common benchmarks, only 0.47 percentage points below the best reported result of 96.29%. These benchmarks are close to saturation, with recent methods clustered within about one percentage point of each other, so this margin reflects a small gap on nearly solved data. A model that overfits to its training source would lose its advantage on data drawn from a different source. Instead, OTSNet ranks first on OST with 79.1%, an occlusion-oriented set whose source is separate from the training data, and stays within 0.47 points of the best result on the common benchmarks, which are also drawn from other sources. This suggests that the robustness of OTSNet generalizes beyond the training source rather than overfitting to it. A higher clean-text accuracy, moreover, does not carry over to challenging text: LPV-B exceeds OTSNet by 0.11 points on the common benchmarks yet trails it by 2.30 points on Union14M-L and 1.4 points on OST. These results show that OTSNet remains competitive across different evaluation benchmarks while achieving strong robustness on challenging text images.

Efficiency–accuracy trade-off: OTSNet has 28.6M parameters and achieves 79.2 FPS (Table 1). Its higher accuracy on Union14M-L and OST relative to SVTRv2 comes with a larger parameter count and lower throughput, as SVTRv2 uses 19.80M parameters and runs at 143 FPS. Compared with LPV-B (30.54M parameters, 82.6 FPS), OTSNet achieves higher accuracy on these challenging benchmarks with fewer parameters and slightly lower throughput.

4.2.1. Effect of DAME

To validate the effectiveness of DAME, we compare the MHSA-only ViT baseline with different attention configurations, as shown in Table 2.

The MHSA-only baseline achieves an average accuracy of 82.89%, whereas replacing all MHSA blocks with DMHA reduces the result to 77.83%. This suggests that using differential attention alone may suppress useful visual responses when global context has not been sufficiently modeled. By interleaving MHSA and DMHA, DAME improves the average accuracy to 83.50%, showing that the two attention mechanisms are more effective when used together.

We further replace DMHA with sparse, background-aware, and contrastive attention under the same Macaron structure. These three variants obtain average accuracies of 83.01%, 82.11%, and 82.71%, respectively, while DAME achieves 83.50%. A similar difference is observed on OST, where DAME reaches 79.1%, compared with 78.2–78.3% for the alternative attention mechanisms. These results indicate that attention differencing provides a more suitable complement to MHSA in the tested encoder.

Table 2
Ablation study on the effectiveness of DAME on Union14M-L and OST.

Var.	MHSA	Cur	M-O	Art	Con	Sal	M-W	Gen	Avg	OST
–	✓	86.2	87.0	76.1	81.0	**83.6**	81.6	84.7	82.89	78.3
D	–	81.0	81.4	69.1	75.6	78.9	76.0	82.8	77.83	76.8
S	–	86.1	86.9	75.2	81.5	82.8	81.0	84.4	82.56	77.9
B	–	85.6	85.8	76.4	81.3	83.0	81.9	84.5	82.64	77.7
C	–	85.7	87.2	73.9	81.4	82.8	81.2	84.2	82.34	77.7
S	✓	86.4	87.2	76.3	81.3	82.4	82.8	84.7	83.01	78.2
B	✓	84.5	86.8	74.2	80.9	82.3	81.6	84.5	82.11	78.3
C	✓	86.4	86.3	75.4	81.3	**83.6**	81.4	84.6	82.71	78.2
D	✓	**87.2**	**87.7**	**76.7**	**81.6**	**83.6**	**82.9**	**84.8**	**83.50**	**79.1**

Var. denotes the alternative attention variant: D, DMHA; S, Sparse (top-k); B, Background-aware (foreground-gated); and C, Contrastive (dual-attention gating).
The first row is the MHSA-only baseline. For D/S/B/C, ✓ in the MHSA column indicates the Macaron structure: MHSA × 2 → Var. × 1 → MHSA × 6 → Var. × 1 → MHSA × 2, where Var. is the attention specified in the first column. "–" indicates that all 12 layers use the specified attention.

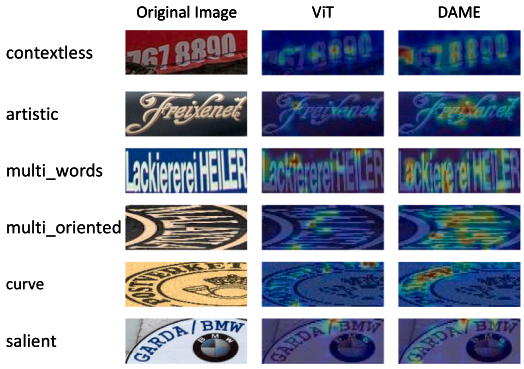


Fig. 6. Comparison of attention maps on Union14M-L images between ViT and DAME. Left: input image; middle: ViT; right: DAME.

Table 3
Ablation study on the impact of λ_{init}.

λ_{init}	Cur	M-O	Art	Con	Sal	M-W	Gen	Avg	OST
0.05	87.2	87.7	76.7	81.6	83.6	82.9	84.8	**83.5**	**79.1**
0.1	87.6	87	76.2	82.8	84.3	81.3	84.8	83.43	78.9
0.15	88.1	88	75.3	81.4	84	80.4	84.7	83.13	78.8

The attention maps in Fig. 6 provide a visual comparison between the MHSA-only ViT and DAME. Compared with ViT, DAME assigns stronger responses to the text regions and shows less activation on the surrounding background.

Fig. 7 further compares DAME with the three alternative attention mechanisms on a cluttered image. DAME retains clearer responses on the character regions while producing lower activation in the background. This observation is consistent with the quantitative results in Table 2.

We further evaluate the impact of λ_{init} in Table 3. The three initialization values of 0.05, 0.10, and 0.15 achieve average accuracies of 83.50%, 83.43%, and 83.13%, respectively. Their corresponding OST accuracies are 79.1%, 78.9%, and 78.8%. The relatively small differences show that DAME is not highly sensitive to λ_{init} within the tested range, although 0.05 gives the best result on both benchmarks. This is because λ_{init} only provides the initial value, while the effective λ is adjusted during training through the learnable parameters λ_{q_1}, λ_{k_1}, λ_{q_2}, and λ_{k_2}.

To examine this adaptation process, we track the effective λ values of the N_2 and N_4 DMHA blocks during training. As shown in Fig. 8, the trajectories starting from different initialization values gradually stabilize as training proceeds. The effective values converge to approximately

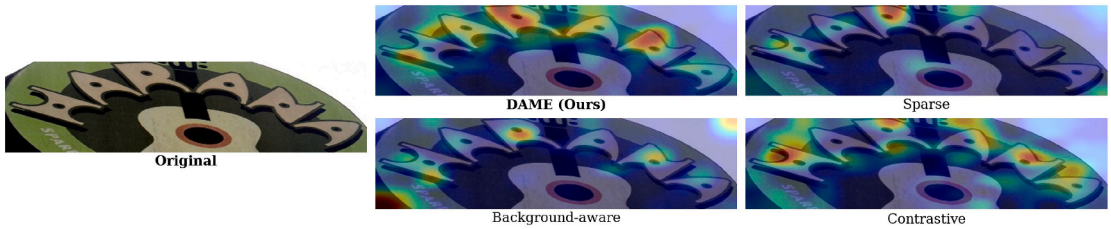


Fig. 7. Attention maps of DAME and alternative attention variants on a cluttered example. Left: input image; right: DAME (Ours), Sparse, Background-aware, and Contrastive.

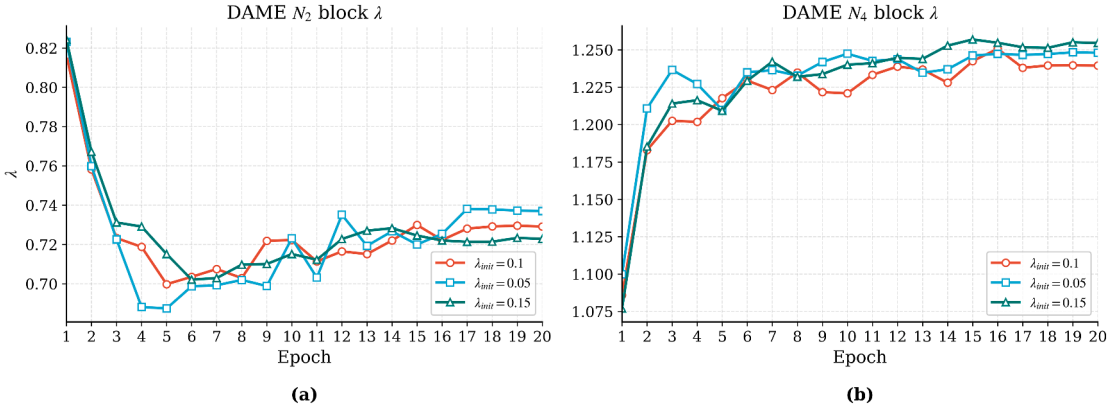


Fig. 8. Convergence trajectories of the dynamic trade-off parameter λ across training epochs for the (a) N_2 and (b) N_4 DMHA blocks.

Table 4
Ablation study on PAM, SQ, and MMCV on Union14M-L and OST.

PAM	MMCV	SQ	Cur	M-O	Art	Con	Sal	M-W	Gen	Avg	OST
✓	–	–	82.8	80.5	69.2	78.3	79.3	73.4	81.9	77.91	71.6
–	✓	–	87.0	87.0	75.3	81.5	83.6	82.0	84.6	83.00	77.3
✓	✓	–	86.0	85.8	73.2	81.4	**83.9**	79.5	84.3	82.01	77.1
✓	✓	✓	**87.2**	**87.7**	**76.7**	**81.6**	83.6	**82.9**	**84.8**	**83.50**	**79.1**

✓: corresponding module is enabled. In the PAM-only configuration, prediction is based on the position-aligned features F_u; in the MMCV-only configuration, F_v is used as visual memory. SQ is evaluated within the PAM–SQ–MMCV pathway.

0.73 in the N_2 block and 1.24 in the N_4 block. This behavior is consistent with the close results obtained under different initialization settings and shows that the attention trade-off is adjusted separately at different layers rather than remaining fixed at its initial value.

4.2.2. Effect of PAM, SQ, and MMCV

We study the effect of combining PAM, SQ, and MMCV, as summarized in Table 4.

PAM alone gives 77.91% average accuracy, whereas MMCV using the visual features directly reaches 83.00%. When the position-aware features produced by PAM are passed to MMCV without SQ, the average accuracy decreases to 82.01%. This suggests that position alignment alone may not provide a suitable intermediate representation for multimodal decoding. After SQ is introduced, the complete PAM–SQ–MMCV pathway achieves 83.50% on Union14M-L and 79.1% on OST. Compared with PAM and MMCV without SQ, the gains are 1.49 and 2.0 percentage points, respectively. The larger gain on OST suggests that the discrete representation is useful when part of the visual evidence is missing. As shown in Fig. 9, SQ also produces more concentrated responses around the character regions in images with noise and background clutter.

4.2.3. Components of SQ and L_{sq}

To further examine SQ, we compare it with several continuous and discrete bottleneck designs in Table 5. A plain MLP improves the Union14M-L average from 82.01% to 82.56%, but decreases the OST accuracy from 77.1% to 73.2%. Adding L_{sq} changes the average accuracy from 82.56% to 82.70%, while increasing the OST accuracy from 73.2% to 78.7%. Thus, in the MLP variant, character-level supervision brings a larger improvement on OST than on the Union14M-L average.

The Linear + Softmax + Embedding variant projects the position-aware features to the vocabulary dimension and uses the resulting probabilities to form a weighted mixture of semantic embeddings. It obtains 83.04% on Union14M-L and 78.6% on OST. SQ increases these results to 83.50% and 79.1%, respectively, with a larger difference on Multi-Word (79.6% vs. 82.9%).

The Stop-Gradient MLP blocks gradients before back-projection and obtains 82.90% on Union14M-L and 78.6% on OST.

To compare SQ directly with Straight-Through VQ, we keep the PAM features, semantic embedding table, L_{sq}, MMCV, model architecture, and training settings unchanged; only the quantization rule is changed. Straight-Through VQ uses deterministic hard $\arg\max$ selection in the forward pass and a Softmax surrogate in back-propagation, whereas SQ applies hard selection to Gumbel-perturbed logits and uses the corresponding continuous distribution for back-propagation. Straight-Through VQ obtains 82.87% on Union14M-L, 78.1% on OST, and 80.3% on Multi-Word, compared with 83.50%, 79.1%, and 82.9% for SQ, respectively. Under this controlled setting, SQ performs better than Straight-Through VQ on both benchmark suites.

Table 5
Ablation study of SQ and alternative bottleneck designs on Union14M-L and OST.

Method	L_{sq}	Cur	M-O	Art	Con	Sal	M-W	Gen	Avg	OST
None	–	86.0	85.8	73.2	81.4	**83.9**	79.5	84.3	82.01	77.1
MLP	–	86.2	87.2	74.8	81.8	83.1	80.3	84.5	82.56	73.2
Linear + Softmax + Emb.	✓	87.0	87.5	**77.4**	**81.9**	83.2	79.6	84.7	83.04	78.6
MLP	✓	85.8	87.7	75.8	81.1	82.8	81.3	84.5	82.70	78.7
Stop-Gradient MLP	✓	86.6	86.9	75.9	81.8	82.6	81.8	84.7	82.90	78.6
Straight-Through VQ	✓	87.0	87.5	75.7	81.0	83.9	80.3	84.7	82.87	78.1
SQ (Ours)	✓	**87.2**	**87.7**	76.7	81.6	83.6	**82.9**	**84.8**	**83.50**	**79.1**

None: baseline without a semantic bottleneck.
MLP: two linear layers mapping features from D to 128 and back to D.
Linear + Softmax + Emb.: Softmax probabilities are used as continuous mixture weights over the semantic vectors.
Stop-Gradient MLP: features are projected from D to C, detached before back-projection, and then mapped back to D.
Straight-Through VQ: deterministic hard $\arg\max$ selection over the same semantic embedding table, with a Softmax surrogate for back-propagation and without Gumbel perturbation.
SQ (Ours) : hard Gumbel-Softmax assignment with a one-hot forward output and a continuous backward surrogate.
✓: L_{sq} is enabled.

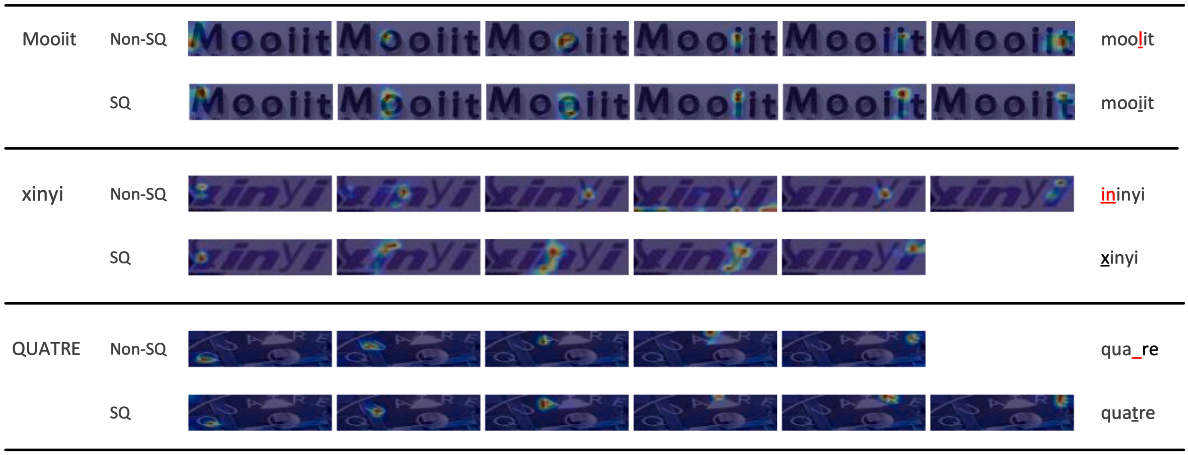


Fig. 9. Visualization of the impact of the SQ module on challenging scene text recognition.

Table 6
Ablation study on the impact of the weighting coefficient α_1 for L_{sq}.

α_1	Cur	M-O	Art	Con	Sal	M-W	Gen	Avg	OST
0.0	86.4	86.0	74.0	80.2	83.1	79.5	84.3	81.93	77.8
0.2	87.4	87.7	75.3	81.8	83.8	79.7	84.7	82.91	77.0
0.3	87.2	87.7	76.7	81.6	83.6	82.9	84.8	**83.50**	**79.1**
0.4	87.2	87.7	75.6	80.9	84.4	81.8	84.8	83.20	78.2

Table 6 further investigates the weighting coefficient α_1 of L_{sq}. Without this loss, the model obtains 81.93% on Union14M-L and 77.8% on OST. The average accuracy improves under all three nonzero settings, although it does not vary monotonically with the loss weight. Among the tested values, $\alpha_1 = 0.3$ gives the highest accuracy on Union14M-L (83.50%) and OST (79.1%). We therefore use $\alpha_1 = 0.3$ in the remaining experiments.

L_{sq} supervises the position-wise logits using the target character at each position. The OST result of the supervised MLP also shows that this intermediate supervision is useful when visual evidence is incomplete. To examine its effect on the learned representation, we project the glyph semantic features of 10 frequently confused character categories into two dimensions using t-SNE. As shown in Fig. 10, the features learned without L_{sq} overlap across several character categories, whereas the supervised features form better-separated clusters.

4.2.4. *DAME block placement and PAM character slots*

To study the placement of DMHA blocks, we compare four DAME layouts in Table 7. All variants contain the same ten MHSA blocks and two DMHA blocks, with only their order changed. The interleaved layout achieves the best average accuracy of 83.5% on Union14M-L. Grouping the two DMHA blocks in the middle or at the end gives similar but lower results, while placing both blocks at the front causes a clear performance drop. These results show that the effect of DMHA depends on its position in the encoder, and that interleaving it with MHSA is more effective among the tested layouts. The interleaved and end-stacked variants both achieve 79.1% on OST.

Fig. 11 compares attention maps across the four layouts. At layer 2, responses in the interleaved, middle-stacked, and end-stacked layouts broadly cover the text regions, whereas the front-stacked layout shows less complete coverage of the character regions and strong responses outside the text. At layer 12, its character coverage remains less complete than that of the interleaved layout. Only the front-stacked model misrecognizes this example. These observations suggest that applying differential attention directly to early features may suppress useful glyph responses along with background responses when glyph and background features are not yet sufficiently distinguished. Two consecutive DMHA blocks at the encoder entrance may therefore hinder the formation of text-focused representations. The other layouts allow MHSA to build

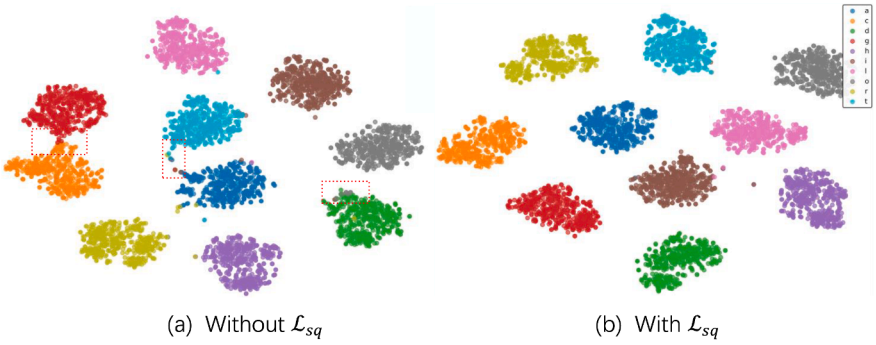


Fig. 10. Visualization of character feature distribution.

Table 7
Ablation study of DAME block placement and PAM character-slot encoding on Union14M-L and OST.

Variant	Cur	M-O	Art	Con	Sal	M-W	Gen	Avg	OST
DAME block placement									
Front-stacked	62.0	56.2	59.8	67.9	60.5	63.1	79.8	64.2	74.3
Middle-stacked	86.0	86.8	72.8	**82.9**	82.9	81.9	84.5	82.5	78.4
End-stacked	85.0	86.1	75.1	81.4	83.1	81.7	84.6	82.4	**79.1**
Interleaved (Ours)	**87.2**	**87.7**	**76.7**	81.6	**83.6**	**82.9**	**84.8**	**83.5**	**79.1**
PAM character-slot encoding									
Learnable embedding	**87.5**	**87.8**	75.7	**82.4**	83.4	81.2	84.6	83.2	78.9
Fixed sinusoidal (Ours)	87.2	87.7	**76.7**	81.6	**83.6**	**82.9**	**84.8**	**83.5**	**79.1**

M and D denote MHSA and DMHA blocks, respectively. The front-stacked, middle-stacked, end-stacked, and interleaved layouts are D^2-M^{10}, M^5-D^2-M^5, M^{10}-D^2, and M^2-D-M^6-D-M^2, respectively. Bold denotes the best result in each ablation group.

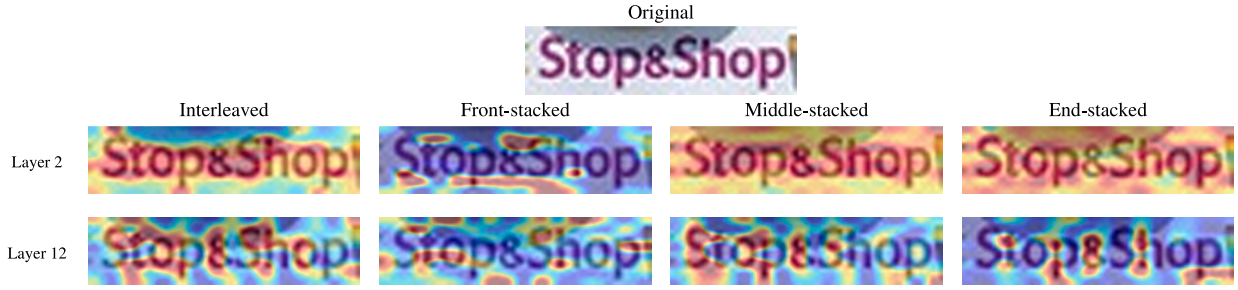


Fig. 11. Attention maps at layers 2 and 12 across four DAME layouts for the same input.

contextual features before differential refinement, which may explain their better performance in the layout ablation.

We further replace the fixed sinusoidal encoding in PAM with learnable character-slot embeddings while keeping the remaining settings unchanged. As shown in Table 7, the fixed encoding achieves average accuracies of 83.5% on Union14M-L and 79.1% on OST, compared with 83.2% and 78.9% for the learnable variant. Although the differences are small and vary across subsets, the fixed sinusoidal encoding gives better aggregate results on both benchmarks. We therefore retain it in PAM.

4.2.5. MMCV masking strategies

To study the proposed MMCV attention mask, we compare four configurations in Table 8. Only memory visibility is changed; the model architecture, training settings, loss functions, and decoding procedure remain unchanged. Each configuration uses its corresponding mask during both training and inference.

The visual-only mask retains full visual access and masks out all semantic slots, achieving 83.01% average accuracy on Union14M-L and 78.9% on OST. Applying a causal mask directly to the concatenated visual and semantic memory produces the joint-causal variant. In this configuration, each query can access only a visual prefix, while all semantic slots remain masked out. Accuracy decreases to 80.93% and 77.8%, respectively. Unlike character-aligned semantic slots, visual tokens represent spatial image features, so restricting them according to the decoding position limits the visual locations available to each query. This comparison shows the benefit of retaining full visual access during decoding.

The full semantic access variant retains full visual access and allows each query to attend to all semantic slots, increasing the Union14M-L average to 83.24%, but decreasing OST accuracy to 78.7% compared with the visual-only mask. Thus, unrestricted semantic access does not consistently improve recognition across the two benchmarks. The proposed MMCV attention mask exposes only the current and preceding semantic slots while keeping the complete visual memory available. This preserves access to visual evidence across the image and organizes semantic access according to the character decoding order. OTSNet achieves 83.50% on Union14M-L and 79.1% on OST, giving the best aggregate results on both benchmarks among the four configurations.

Table 8
Ablation study of MMCV masking strategies on Union14M-L and OST.

Masking strategy	Cur	M-O	Art	Con	Sal	M-W	Gen	Avg	OST
Visual-only	86.8	87.1	76.0	81.1	83.0	82.6	84.5	83.01	78.9
Full semantic access	**87.2**	87.3	76.3	**82.0**	83.3	81.9	84.7	83.24	78.7
Joint causal	84.2	83.3	73.9	78.2	82.1	81.0	83.8	80.93	77.8
OTSNet (Ours)	**87.2**	**87.7**	**76.7**	81.6	**83.6**	**82.9**	**84.8**	**83.50**	**79.1**

Bold denotes the best result in each column. Avg denotes the average accuracy across the seven Union14M-L subsets.



Fig. 12. Representative failure cases of OTSNet on the six common benchmarks, Union14M-L, and OST. GT and Pred. denote the ground-truth transcription and model prediction, respectively.

4.2.6. *Failure analysis*

Fig. 12 presents representative failure cases of OTSNet. Most errors involve substitutions between visually similar characters when the text is blurred, occluded, rotated, or written in unusual fonts. The examples also include missing spaces in multi-word text and incorrect characters in narrow or heavily distorted images. Several failures occur in short words, where ambiguity in a single glyph is sufficient to change the final prediction. These cases show that OTSNet still has difficulty when the available visual evidence cannot clearly distinguish character shapes.

4.3. *Limitations*

Although OTSNet remains competitive on the common benchmarks, its average accuracy of 95.82% is 0.47 percentage points below the best reported result of 96.29%. This gap cannot be attributed solely to the absence of a standalone language model. MMCV already exploits autoregressive character context during decoding, and the methods that surpass OTSNet on the common benchmarks also differ in their visual encoders, decoders, and pre-training. Three of the four methods above OTSNet, namely ABINet, MATRN, and BUSNet, rely on additional pre-training; the only pre-training-free method among them, LPV-B, exceeds OTSNet by just 0.11 points on the common benchmarks while remaining less robust on the challenging suites (Section 4.2). A separate language model may therefore yield a small clean-text gain, but it is neither necessary nor sufficient to close this gap. A constructive direction is to attach a lightweight, decoupled language model in the manner of ABINet (Fang et al., 2021) as a post-processing re-scoring module after training, leaving the visual backbone and the MMCV decoder unchanged so that the from-scratch, gradient-blocking-free training remains intact. The key constraint is that such re-scoring must defer to glyph-level evidence and must not override visual predictions on degraded images or non-lexical strings. We leave this constrained integration to future work. In addition, autoregressive decoding introduces sequential inference cost. Our experiments are also limited to English STR benchmarks, leaving multilingual recognition, longer text sequences, and larger character vocabularies for further study.

5. Conclusions

We presented OTSNet, an end-to-end STR framework organized into Observation, Thinking, and Spelling stages. DAME alternates MHSA and DMHA to extract visual features under background interference, while PAM aligns these features with fixed character positions. SQ uses hard Gumbel-Softmax to select one glyph-level semantic unit for each position-aware feature, and MMCV combines these units with visual and character features for autoregressive decoding. Experiments show that OTSNet achieves 83.5% on Union14M-L and 79.1% on OST, while remaining competitive on the six common benchmarks.

CRediT authorship contribution statement

Lixu Sun: Writing – original draft, Software, Methodology, Conceptualization; **Jian Peng:** Writing – original draft, Visualization; **Nurmemet Yolwas:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition; **Wushour Silamu:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Key Research and Development Program of China under Grant No. 2023B01005, titled *Research and Application of Key Technologies for Intelligent Perception, Analysis, and Decision-Making in the Multimodal Internet Content Security Ecosystem*.

Data availability

The ICDAR 2013 (IC13) (Karatzas et al., 2013), Street View Text (SVT) (Wang et al., 2011), IIIT5K Word (IIIT5K) (Mishra et al., 2012), ICDAR 2015 (IC15) (Karatzas et al., 2015), Street View Text Perspective (SVTP) (Phan et al., 2013), CUTE80 (Risnumawan et al., 2014), the Union14M-L dataset (Jiang et al., 2023), and Occluded Scene Text (OST) (Wang et al., 2021) are publicly available.

Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.neunet.2026.109829.

References

Atienza, R. (2021). Vision transformer for fast and efficient scene text recognition. In *International conference on document analysis and recognition* (pp. 319–334). Springer.

Bautista, D., & Atienza, R. (2022). Scene text recognition with permuted autoregressive sequence models. In *European conference on computer vision* (pp. 178–196). Springer.

Bigverdi, M., Luo, Z., Hsieh, C.-Y., Shen, E., Chen, D., Shapiro, L. G., & Krishna, R. (2025). Perception tokens enhance visual reasoning in multimodal language models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 3836–3845). https://doi.org/10.1109/CVPR52734.2025.00363

Cheng, C., Wang, P., Da, C., Zheng, Q., & Yao, C. (2023). Lister: Neighbor decoding for length-insensitive scene text recognition. In *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)* (pp. 19541–19551).

Cheng, Z., Xu, Y., Bai, F., Niu, Y., Pu, S., & Zhou, S. (2018). AON: Towards arbitrarily-oriented text recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 5571–5579).

Du, Y., Chen, Z., Jia, C., Yin, X., Zheng, T., Li, C., Du, Y., & Jiang, Y.-G. (2022). SVTR: Scene text recognition with a single visual model. In L. D.Raedt (Ed.), *Proceedings of the thirty-first international joint conference on artificial intelligence, IJCAI-22* (pp. 884–890). International Joint Conferences on Artificial Intelligence Organization. Main Track https://doi.org/10.24963/ijcai.2022/124

Du, Y., Chen, Z., Xie, H., Jia, C., & Jiang, Y.-G. (2025). SVTRv2: CTC beats encoder-decoder models in scene text recognition. In *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)* (pp. 20147–20156).

Fang, S., Xie, H., Wang, Y., Mao, Z., & Zhang, Y. (2021). Read like humans: Autonomous, bidirectional and iterative language modeling for scene text recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 7098–7107).

Fu, Z., Xie, H., Fang, S., Wang, Y., Xing, M., & Zhang, Y. (2023). Learning pixel affinity pyramid for arbitrary-shaped text detection. *ACM Transactions on Multimedia Computing, Communications and Applications*, *19*(1s), 1–24.

Gupta, A., Dar, G., Goodman, S., Ciprut, D., & Berant, J. (2021). Memory-efficient transformers via top-k attention. In *Proceedings of the second workshop on simple and efficient natural language processing* (pp. 39–52).

Jang, E., Gu, S., & Poole, B. (2017). Categorical reparameterization with Gumbel-Softmax. In *International conference on learning representations*.

Jiang, Q., Wang, J., Peng, D., Liu, C., & Jin, L. (2023). Revisiting scene text recognition: A data perspective. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 20543–20554).

Karatzas, D., Gomez-Bigorda, L., Nicolaou, A., Ghosh, S., Bagdanov, A., Iwamura, M., Matas, J., Neumann, L., Chandrasekhar, V. R., Lu, S. et al. (2015). ICDAR 2015 competition on robust reading. In *2015 13th international conference on document analysis and recognition (ICDAR)* (pp. 1156–1160). IEEE.

Karatzas, D., Shafait, F., Uchida, S., Iwamura, M., i Bigorda, L. G., Mestre, S. R., Mas, J., Mota, D. F., Almazàn, J. A., & De Las Heras, L. P. (2013). ICDAR 2013 robust reading competition. In *2013 12th international conference on document analysis and recognition* (pp. 1484–1493). IEEE.

Kastner, S., & Ungerleider, L. G. (2000). Mechanisms of visual attention in the human cortex. *Annual Review of Neuroscience*, *23*(1), 315–341.

Lee, J., Park, S., Baek, J., Oh, S. J., Kim, S., & Lee, H. (2020). On recognizing texts of arbitrary shapes with 2D self-attention. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops* (pp. 546–547).

Li, J.-N., Liu, X.-Q., Luo, X., & Xu, X.-S. (2024). Volter: Visual collaboration and dual-stream fusion for scene text recognition. *IEEE Transactions on Multimedia*, *26*, 6437–6448.

Li, M., Fu, B., Zhang, Z., & Qiao, Y. (2021). Character-aware sampling and rectification for scene text recognition. *IEEE Transactions on Multimedia*, *25*, 649–661.

Luo, C., Lin, Q., Liu, Y., Jin, L., & Shen, C. (2021). Separating content from style using adversarial learning for recognizing text in the wild. *International Journal of Computer Vision*, *129*, 960–976.

Mishra, A., Alahari, K., & Jawahar, C. V. (2012). Scene text recognition using higher order language priors. In *BMVC-British machine vision conference*. BMVA.

Na, B., Kim, Y., & Park, S. (2022). Multi-modal text recognition networks: Interactive enhancements between visual and semantic features. In *European conference on computer vision* (pp. 446–463). Springer.

Ozkan, S., Maracani, A., Ozay, M., Kim, H., Cho, S., Noh, E., Min, J., & Cho, J. M. (2025). Efficient and accurate scene text recognition with cascaded-transformers. In *Proceedings of the 16th ACM multimedia systems conference* (pp. 335–341). https://doi.org/10.1145/3712676.3719267

Phan, T. Q., Shivakumara, P., Tian, S., & Tan, C. L. (2013). Recognizing text with perspective distortion in natural scenes. In *Proceedings of the IEEE international conference on computer vision* (pp. 569–576).

Qiao, Z., Zhou, Y., Wei, J., Wang, W., Zhang, Y., Jiang, N., Wang, H., & Wang, W. (2021). PIMNet: A parallel, iterative and mimicking network for scene text recognition. In *Proceedings of the 29th ACM international conference on multimedia* (pp. 2046–2055).

Qiao, Z., Zhou, Y., Yang, D., Zhou, Y., & Wang, W. (2020). SEED: Semantics enhanced encoder-decoder framework for scene text recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 13528–13537).

Qu, L., Zhang, H., Liu, Y., Wang, X., Jiang, Y., Gao, Y., Ye, H., Du, D. K., Yuan, Z., & Wu, X. (2025). TokenFlow: Unified image tokenizer for multimodal understanding and generation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 2545–2555). https://doi.org/10.1109/CVPR52734.2025.00243

Risnumawan, A., Shivakumara, P., Chan, C. S., & Tan, C. L. (2014). A robust arbitrary text detection system for natural scene images. *Expert Systems with Applications*, *41*(18), 8027–8048.

Sheng, F., Chen, Z., & Xu, B. (2019). NRTR: A no-recurrence sequence-to-sequence model for scene text recognition. In *2019 International conference on document analysis and recognition (ICDAR)* (pp. 781–786). IEEE.

Shi, B., Bai, X., & Yao, C. (2016). An end-to-end trainable neural network for image-based sequence recognition and its application to scene text recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *39*(11), 2298–2304.

Shi, B., Yang, M., Wang, X., Lyu, P., Yao, C., & Bai, X. (2018). Aster: An attentional scene text recognizer with flexible rectification. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *41*(9), 2035–2048.

van den Oord, A., Vinyals, O., & Kavukcuoglu, K. (2017). Neural discrete representation learning. In *Advances in neural information processing systems* (pp. 6306–6315). (*vol. 30*).

Wang, K., Babenko, B., & Belongie, S. (2011). End-to-end scene text recognition. In *2011 International conference on computer vision* (pp. 1457–1464). IEEE.

Wang, P., Da, C., & Yao, C. (2022). Multi-granularity prediction for scene text recognition. In *European conference on computer vision* (pp. 339–355). Springer.

Wang, S., Wang, Y., Qin, X., Zhao, Q., & Tang, Z. (2019). Scene text recognition via gated cascade attention. In *2019 IEEE International conference on multimedia and expo (ICME)* (pp. 1018–1023). https://doi.org/10.1109/ICME.2019.00179

Wang, T., Zhu, Y., Jin, L., Luo, C., Chen, X., Wu, Y., Wang, Q., & Cai, M. (2020). Decoupled attention network for text recognition. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 12216–12224). (*vol. 34*).

Wang, Y., Xie, H., Fang, S., Wang, J., Zhu, S., & Zhang, Y. (2021). From two to one: A new scene text recognizer with visual language modeling network. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 14194-14203).

Wei, J., Zhan, H., Lu, Y., Tu, X., Yin, B., Liu, C., & Pal, U. (2024). Image as a language: Revisiting scene text recognition via balanced, unified and synchronized vision-language reasoning network. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 5885–5893). (*vol. 38*).

Wu, L., Xu, Y., Hou, J., Chen, C. L. P., & Liu, C.-L. (2022). A two-level rectification attention network for scene text recognition. *IEEE Transactions on Multimedia*, *25*, 2404–2414.

Xu, J., Wang, Y., Xie, H., & Zhang, Y. (2024). OTE: Exploring accurate scene text recognition using one token. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 28327–28336).

Yang, M., Yang, B., Liao, M., Zhu, Y., & Bai, X. (2024). Class-aware mask-guided feature refinement for scene text recognition. *Pattern Recognition*, *149*, 110244.

Ye, T., Dong, L., Xia, Y., Sun, Y., Zhu, Y., Huang, G., & Wei, F. (2024). Differential transformer. arXiv:2410.05258

Yu, D., Li, X., Zhang, C., Liu, T., Han, J., Liu, J., & Ding, E. (2020). Towards accurate scene text recognition with semantic reasoning networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 12113–12122).

Yue, X., Kuang, Z., Lin, C., Sun, H., & Zhang, W. (2020). RobustScanner: Dynamically enhancing positional clues for robust text recognition. In *European conference on computer vision* (pp. 135–151). Springer.

Zhang, B., Xie, H., Wang, Y., Xu, J., & Zhang, Y. (2023). Linguistic more: Taking a further step toward efficient and accurate scene text recognition. In E. Elkind (Ed.), *Proceedings of the thirty-second international joint conference on artificial intelligence, IJCAI-23* (pp. 1704–1712). International Joint Conferences on Artificial Intelligence Organization. Main Track https://doi.org/10.24963/ijcai.2023/189

Zhang, Y., Liu, C., Wei, J., Yang, X., Zhou, Y., Ma, C., & Ji, X. (2025). Linguistics-aware masked image modeling for self-supervised scene text recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 9318–9328). https://doi.org/10.1109/CVPR52734.2025.00870

Zheng, T., Chen, Z., Fang, S., Xie, H., & Jiang, Y.-G. (2024). CDistNet: Perceiving multi-domain character distance for robust text recognition. *International Journal of Computer Vision*, *132*(2), 300–318.

Zhou, J., Wang, X., Dai, J., & Han, J. (2023). A decoupled language model based on contrastive attention mechanism for scene text recognition. In *Proceedings of the 2023 9th international conference on computing and artificial intelligence* ICCAI '23 (pp. 452–458). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3594315.3594356